ID: 221-15-5428

Assignment

Answer to the que: NO: 01

(a).

(a)

Here Given,

IP address: 172.16.0.0/16

Subnet mask for HR dept:

each customers needs 256 IP addressed

$256 = 2^8$

So, required host bit = 8

∴ Number of Network bits = 32 − 8 = 24

∴ Subnet Mask: 11111111.11111111.11111111.00000000

= 255.255.255.0

Subnet Mask for Finance dept:

each customer needs 128 ip addressed

$128 = 2^7$

∴ Network bits = 32 − 7 = 25

So, Subnet Mask: 255.255.255.128

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Subnet Mask for IT dept:

each customer needs 128 ip addressed

$128 = 2^7$

$\therefore$ Network bits $= 32 - 7 = 25$

$\therefore$ subnet Mask $= 255.255.255.128$

(b)

Subblocks for each dept:

HR dept:

Range: 0.0.0.255

N.A for customer 1: 172.16.0.0/24

B.A for customer 1: 172.16.0.255/24

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N.A for customer 64: 172.16.63.0/24

B.A for customer 64: 172.16.63.255/24

$\therefore$ Range of subnets: 172.16.0.0/24 to 172.16.63.0/24

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Finance dept:

Range: 0.0.0.127

N.A for customer 1: 172.16.64.0/25

B.A for customer 1: 172.16.64.127/25

- - - -

B.A for customer 128: 172.16.191.255/25

∴ Range for subnets: 172.16.128.0/25 to

172.16.191.128/25

(c)

HR dept:

Total address used = 64 × 256

= 16,384 addresses

Finance dept:

Total address used = 128 × 128

= 16,384 addresses

IT dept:

Total address used = 128 × 128

= 16,384 addresses

So, Total ip used in all department.

= 3 × 16,384 = 49,152 addresses.

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